

11.4 部分习题参考简答

$$1. \quad (1) \quad \frac{\partial(u, v)}{\partial(x, y)} = \begin{pmatrix} \frac{x}{\sqrt{x^2 + y^2}} & \frac{y}{\sqrt{x^2 + y^2}} \\ -\frac{y}{x^2 + y^2} & \frac{x}{x^2 + y^2} \end{pmatrix};$$

$$(2) \quad \frac{\partial(u, v)}{\partial(x, y)} = \begin{pmatrix} e^x \cos y & -e^x \sin y \\ e^x \sin y & e^x \cos y \end{pmatrix};$$

$$(3) \quad \frac{\partial(u, v)}{\partial(x, y)} = \begin{pmatrix} \frac{y^2 - x^2}{(x^2 + y^2)^2} & \frac{-2xy}{(x^2 + y^2)^2} \\ \frac{-2xy}{(x^2 + y^2)^2} & \frac{x^2 - y^2}{(x^2 + y^2)^2} \end{pmatrix};$$

$$(4) \quad \frac{\partial(u, v)}{\partial(x, y)} = \begin{pmatrix} \frac{x}{x^2 + y^2} & \frac{y}{x^2 + y^2} \\ \frac{-y}{x^2 + y^2} & \frac{x}{x^2 + y^2} \end{pmatrix}.$$

$$2. \quad \frac{\partial(x, y, z)}{\partial(r, \theta, \varphi)} = \begin{pmatrix} \sin \theta \cos \varphi & r \cos \theta \cos \varphi & -r \sin \theta \sin \varphi \\ \cos \theta \cos \varphi & -r \sin \theta \cos \varphi & -r \cos \theta \sin \varphi \\ \sin \varphi & 0 & r \cos \varphi \end{pmatrix}.$$

$$3. \quad (1) \quad D(f \circ g) = \begin{pmatrix} \frac{y^2 - 2xy - x^2}{(x^2 + y^2)^2} & \frac{x^2 - 2xy - y^2}{(x^2 + y^2)^2} \\ \frac{y^3 - 3x^2y}{(x^2 + y^2)^3} & \frac{x^3 - 3xy^2}{(x^2 + y^2)^3} \\ -\frac{y^3 + x^2y}{x^4 + x^2y^2} & \frac{x^2 + y^2}{x^3 + xy^2} \end{pmatrix};$$

$$d\mathbf{y} = D(f \circ g) d\mathbf{x} = \begin{pmatrix} \frac{y^2 - 2xy - x^2}{(x^2 + y^2)^2} dx + \frac{x^2 - 2xy - y^2}{(x^2 + y^2)^2} dy \\ \frac{y^3 - 3x^2y}{(x^2 + y^2)^3} dx + \frac{x^3 - 3xy^2}{(x^2 + y^2)^3} dy \\ -\frac{y^3 + x^2y}{x^4 + x^2y^2} dx + \frac{x^2 + y^2}{x^3 + xy^2} dy \end{pmatrix}.$$

$$(2) \quad D(f \circ g) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

$$d\mathbf{y} = D(f \circ g) d\mathbf{x} = \begin{pmatrix} dx \\ -dy \end{pmatrix}.$$

$$4. \quad D\mathbf{f} = \frac{\partial(y_1, y_2)}{\partial(x_1, x_2, x_3)} = \begin{pmatrix} x_2 + x_3 & x_1 + x_3 & x_1 + x_2 \\ x_2 x_3 & x_1 x_3 & x_1 x_2 \end{pmatrix},$$

$$d\mathbf{f}(1, -1, -2) = D\mathbf{f}(1, -1, -2) d\mathbf{x} = \begin{pmatrix} -3dx_1 - dx_2 \\ 2dx_1 - 2dx_2 - dx_3 \end{pmatrix}.$$

$$5. \quad df(\mathbf{X}_0) = \begin{pmatrix} 2dx_{11} & dx_{12} \\ dx_{21} & 0 \end{pmatrix}.$$